CPE 4810: Senior Design Project 1, Spring 2026

Concept Proposal Document

Team: Short Circuit

Team Members: Ethan Karnes, Elijah Epps, Justin Turner, Jesse Garcia

Date: February 27th, 2026

Electrical and Computer Engineering

Kennesaw State University

Faculty: Dr. Jeffrey L Yiin

1. The Team

Team Name and Logo

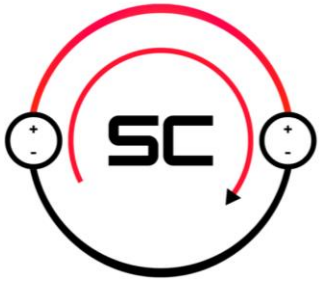
Our team's name is Short Circuit, but we originally went by Memory Overflow. At the time, Memory Overflow was just a temporary choice we came up with at the last minute. Even so, the name captured the idea we wanted to convey (referencing hardware and computer errors). The logo we used was AI-generated and felt sloppy. We didn't feel comfortable using it as our team logo.

We knew we needed to lock in our identity. It became clear that we needed a name and logo that felt intentional and unique. That is how we arrived at Short Circuit, along with a new logo that better reflects our vision and energy.

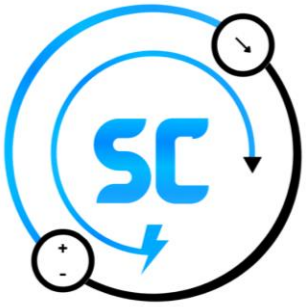
Team Logos:

Original AI generated Logo

Initial Short Circuit Logo



Final Short Circuit Logo



Team Mission and Objective

Our team's mission and objective are to utilize what we have learned in the past few years to create a product that can help others solve their problems.

Team Roles

Justin Turner: Creative Director

Jesse Garcia: Hardware Manager

Elijah Epps: Network Specialist

Ethan Karnes: Lead Market Researcher

2. Our ideas

Main Idea: Life Loop – A Wellness Network

- Customer – Our ideal customers for this device are adults aged 40 to 60 with elderly parents.
- Problem – There is currently no simple way to monitor the wellbeing of elderly people; families often rely on delayed communication methods to check/monitor their loved ones.
- Solution – Our solution is a wearable device that utilizes sensors such as an accelerometer that detects falls or other impacts, GPS to track location, and heart rate sensors to ensure that the wearer isn't experiencing any cardiac irregularities.
- Secret Sauce – Our solution will create a shared safety network through wearable monitoring (accelerometer, GPS, and heart rate sensors) to ensure that the wearer's wellbeing is good. Using the sensors, the wearable will be able to provide crucial data to the family members such as location, time, and nature of emergency. In an event of emergency, the device will notify the wearer's family first, and if there is no response/resolution after a period, it will proceed to contact emergency services.

Secondary Idea: Car Impact Assistant

- Customer – Customers who live in urban areas, parents of teen drivers, and fleet managers
- Problem – Modern vehicles insulate drivers from the road as they mask the feedback required to detect mechanical damage. When drivers hit either a pothole or curb, they are often left guessing whether or not the impact was minor or severe. In other words, there is a lack of objective data when it comes to diagnosing these impacts.
- Solution – Our solution is an embedded system that can quantify the mechanical shock of the impacts in real time. The device would track chassis vibration to distinguish between normal, harmless road conditions and damaging forces. If the device deems the impact to be severe, a dashboard display would provide the driver with an immediate warning to pull over.
- Secret Sauce - The key differentiator of this device would be the system's intelligent impact discrimination. This differs from basic vibration alarms that often trigger every bump (speed bumps, speed cushions, etc), the device would feature advanced signal analysis that would identify the specific signature of a hit. It would then automatically distinguish between the gradual motion of a

speed bump as opposed to the sharp and damaging spike of either a pothole or a curb, ensuring that the driver is only alerted when damage is detected.

3. Customer Validation

Our chosen main product is Life Loop. We believe that this idea will be the best moving forward. After choosing our main product, we needed to do some customer research and validation. We first listed our key assumptions as:

1. Parents are not always easily accessible through the phone
2. Parents are willing to wear a bracelet-sized device constantly
3. Current technology can accurately detect falls, location, and heart rate
4. Customers are willing to trade privacy for increased well being
5. Customers are willing to use app and ecosystem
6. Wearers of the device do not already wear a watch/ bracelet on their wrist

After this, we needed to formulate some questions for our customer outreach interviews. Our group met a few times outside of class and came up with a list of questions that would not only test our assumptions, but expound upon our idea and see if what we wanted to create was worthwhile. Below is our script of questions we asked our interviewees.

Customer Interview Script:

Part 1: (Testing Assumption)

- How often do you worry about a family member or friend?
- Tell me about the last time you tried to reach someone and they didn't answer. What went through your mind?
- How do you currently check on family?
- Walk me through exactly what you do when you want to check on them. Do you text first? Call?
- What is the part you don’t like about checking on family?
- Have you ever avoided checking on someone because you didn't want to get stuck on a long phone call?
- How does your family member currently react when you check in 'too much'? Do they ever express that they feel babysat or nagged?
- If an emergency happened right now (like a fall), how would you find out? Walk me through the realistic timeline of who would call you and when.

- Have you tried using current solutions currently available (Find My, Life360), and if so, how was your experience using them?

Part 2: (Product Design)

We showed the interviewee the stimulus of the initial design.

Part 3: (Fit and Constrains)

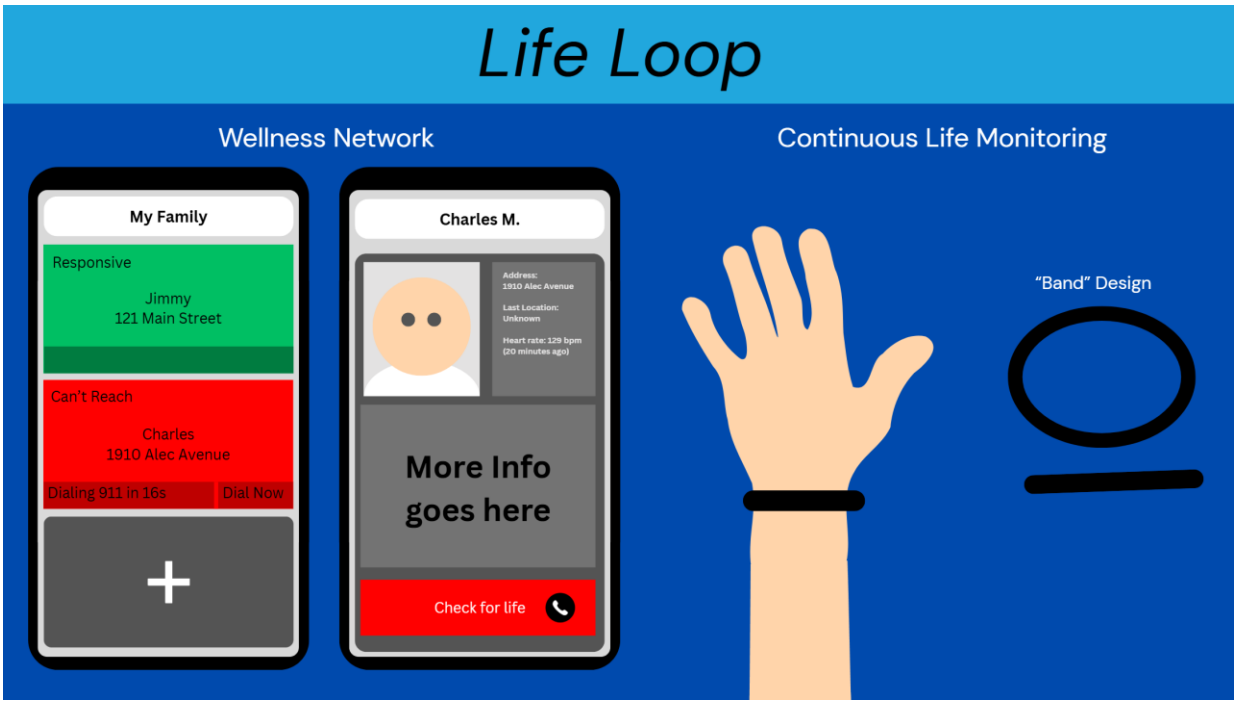
- Do your parents currently wear a watch or Fitbit? If they stopped wearing one in the past, why did they stop?
- What are your criteria that makes a wearable comfortable for everyday use? Does this change with age?
- How comfortable are you with sharing your parent's location and health data?
- Do you currently share your and or your parents location on Google Maps or Life360? If yes, do you ever turn it off?
- How do you and your parents normally handle charging for devices that have batteries/ any wearables?
- What is the most annoying device you have to charge right now? How often does it die?

Part 4: (Feedback)

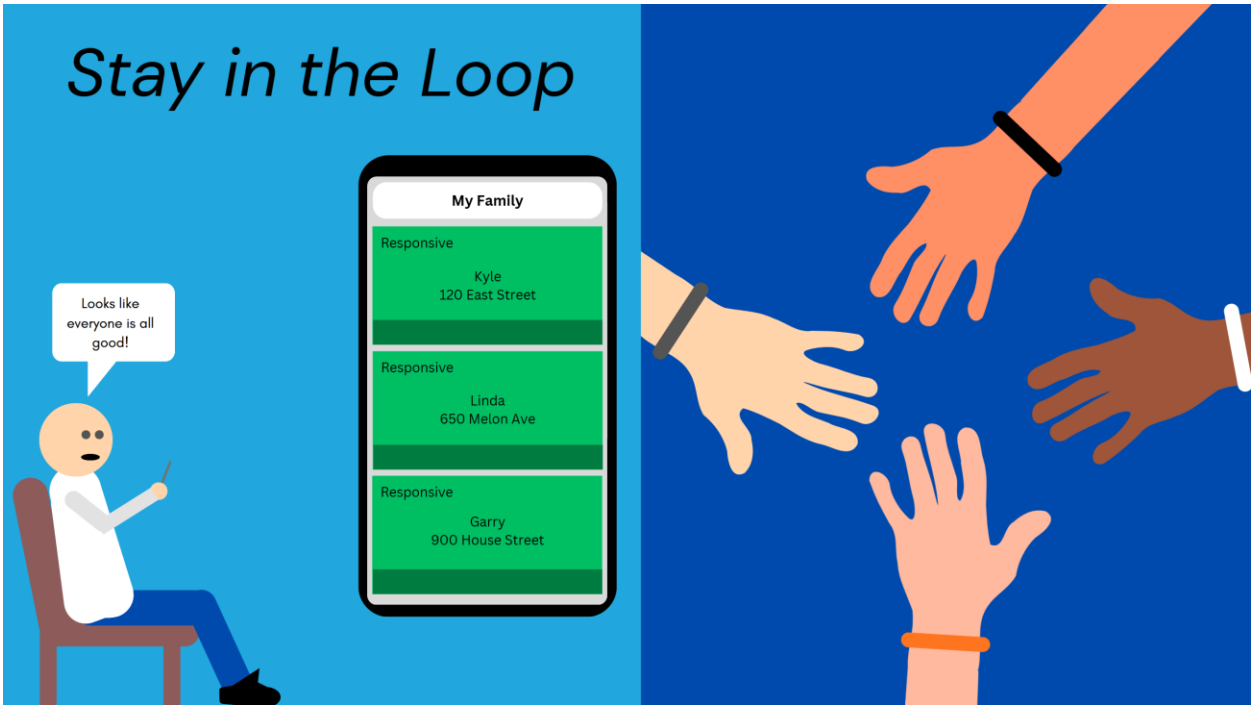
- What other functions would you need the device to accomplish for it to be useful?
- Is there any part/function of the device that seems unnecessary? If yes, please explain why it feels unnecessary.
- What do you think of the design and overall feel of this device? Are there any other features you'd like to see?
- Do you think your parents would use this device? Why/ why not and what is the limiting factor?

With that set of questions, we needed to create an actual mockup to visualize what we are trying to make. After some initial sketches and drawings, we landed on the design that can be seen below.

Life Loop Customer Stimulus



Life Loop Billboard



While thinking about our entire possible customer base, we thought of many different customer profiles but landed on two main groups of customers. Our first main group is the elderly people themselves. We had these as our demographics:

- 65+ in age
- Either living alone or with a partner

Our second main group is adult children with elderly parents:

- 35-55 age range
- Have at least one parent that they worry about

With our interview script completed and our target customer groups, we then set out on the interview process. This first began with the *Customer Interview Plan* assignment and thinking of people we knew. After we all found some people to interview, we started the interview process. Most of our interviews were conducted with the second main group, as we agreed that elderly people usually do not purchase this type of device on their own.

Post Interview Data Analysis

After we interviewed our potential customers and wearers of the product, we started to review the feedback. Across the data we gathered an overwhelming majority of our assumptions were validated. From the list of assumptions above, 1-5 were validated with many of our customers saying that they would buy this type of device for their parents.

A few major points that we heard many times during interviews were that of the physical needs of the device. Mainly, we heard that the device needed to be comfortable, subtle, and accessible. This validated our early design of a simple "band" design where we tried to keep the device as simple as possible but did make us think about how simplicity might be better than to add a ton of features. A few of the interviewees suggested a more rugged design, as their parents were still working or were more physical. We are

One key assumption that was invalidated was number 6 on the list above. We found that after tallying the data, about half of those surveyed had parents that wore something on their wrist already. This invalidated assumption leads us to value the physical nature of the device even more because now we know that we might be "fighting" for wrist space. We also consider that the customers' parents might wear something on only one wrist, leaving the opposite wrist open.

Another point that was heard a few times during interviews was that of data privacy. Some customers were concerned about the data being collected and this point, although less thought about, totally resonated with us. Despite it being closer to a feasibility issue,

we still discussed privacy and how it is a very real unspoken “feature” of any device. This was a somewhat unexpected point that our potential customers brought up and it will shape our implementation should we move forward.

Another unexpected insight we discovered was that many people we spoke with were interested in fitness-related features, such as tracking calories and movement. We initially overlooked this because our focus was on wellness-checking features. However, this is something we can consider including as we move forward with the product.

Our top feedback we got was to keep the device comfortable, cost-effective, and usable. Eleven out of the twelve people we interviewed said that a top concern with them would be comfort. The next biggest concern was battery life, which is something that leans into the feasibility of the device but we still found very valuable as feedback. Lastly, cost was brought up quite a lot during the interviews and is something that we have noted down. Potential customers want a cost-effective solution and voiced that clearly.

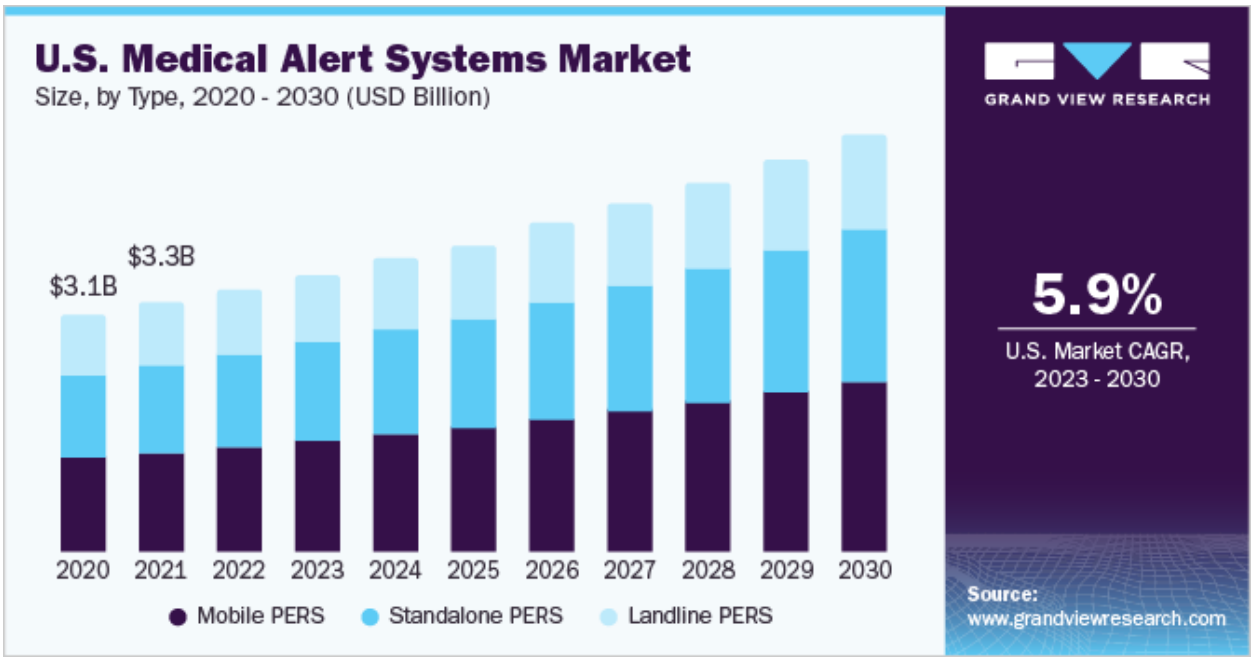
Overall, our customers often worry about their aging parents, but there are few simple, affordable ways to check on their well-being without constant calls. Most current products are either too complex or expensive. Our goal is to create an easy-to-use and affordable solution that allows people to monitor their loved ones' well-being without hassle or high costs.

We feel that our idea has a potential customer base based off of interviews. A lot of our assumptions were validated, with the ones invalidated leading to valuable thinking points.

4. Market Analysis and Competitive Landscape

Upon doing further research, we discovered that the market around geriatric care devices was valued at around $14.80 billion in 2025 and is only expected to grow further. This market is currently projected to reach roughly $26.83 billion by the year 2033 with a current projected annual growth rate of 7.74%. Furthermore, North America seems to be at the helm of this market since it accounted for approximately 40 million wearable devices used yearly.

Diving more into specifics, the medical alert system devices were at an evaluation of $3.3 billion in 2021. With a 5.9% annual growth rate, the evaluation by 2030 is $5.5 billion. As shown in the graph below, the mobile personal emergency response systems is about 1/3 of this market, meaning that in 2030 it would be valued at $1.83 billion alone. This shows us that the market is absolutely there for these types of devices.



Existing Competition

There are multiple devices currently available on the market. Our main competition comes from both general consumer technology and medical alert systems. Below is a list of common devices, strengths, and weaknesses.

- Adaptative Medical Watches (Bay Alarm Medical SOS Smartwatch, etc.)
    - Strength: Fall detection utilizes artificial intelligence to learn the wearer's movements to reduce the chances of a false positive, and they have a monitoring center that supports over 170 languages.
    - Weakness: The battery life on these devices could be better as they only last about 6 to 18 hours on a full charge. Users of these wearables would have to remember to charge them daily to actually use it.
- Premium Wearables (Medical Guardian MGMove, etc):
    - Strength: Sleek and premium aesthetic which is designed in such a way to not look like a medical device. They often boast about their rapid emergency response times, averaging around 29 seconds.
    - Weakness: These devices can be quite expensive as the equipment itself is $199.95 along with a monthly subscription charge of $42.95, and if you want fall detection there is another $10 per month charge. There have also been tests conducted on these wearables in which the fall detection was inaccurate and would record false positives.
- Consumer Smartwatches (Apple Watch, Galaxy Watch, etc.):

- Strength: These smartwatches often look sleek, low-profile, and effectively eliminate the psychological stigma of wearing a medical device. They also offer elaborate health data without paying a subscription fee.
- Weakness: The fall detection algorithms in the smartwatches are usually optimized in a way where only high impact falls are detected, but the soft falls that are common amongst the elderly aren't detected. These wearable devices are made with complex screen navigation, which can be difficult for elderly people to use.

Our Competitive Edge:

Life Loop successfully closes the gap between these rivals. We will be able to precisely detect both huge kinetic impacts and the slow, subtle "soft falls" that are frequently missed by generic smartwatches by incorporating sophisticated algorithms into our accelerometer. Life Loop has a clever, tiered escalation procedure in contrast to the Apple Watch, which routes warnings straight to 911 and may result in enormous, needless ambulance expenses for trivial situations.

The device automatically provides vital information such as location, time, and the type of emergency to the wearer's shared family safety network first in the case of an emergency. After a predetermined amount of time, the gadget will automatically contact emergency services if the family does not respond or resolve the issue. Help is contacted even if the user is incapacitated thanks to this fully automated approach that requires no human participation or button pressing. Furthermore, we guarantee a significantly longer battery life by concentrating only on necessary passive sensors rather than power-hungry wristwatch programs. Lastly, we place a high value on a discreet, fashionable design. By blending in perfectly, the user avoids the significant social stigma attached to conventional medical buttons, guaranteeing that their location tracking and health monitoring stay completely private.

5. Concluding Remarks

What We Learned

During the first phase of the class, our team learned to approach ideation with an entrepreneurial mindset. We discovered the importance of identifying and understanding a solid customer base. We also recognized the value of feedback. Interviewing customers and testing our assumptions helped guide and improve our product design. Most importantly, we learned how to work as a team and collaborate.

Feedback for Dr. Yiin

Dr. Yiin provides thoughtful guidance and solid advice without ever forcing his perspective on the team. His insights help narrow the project's scope and support informed decision-making, while still leaving room for the team to form its own opinions. His feedback has been extremely helpful in keeping the project on the right track.

What's next for the team

As a team, we plan to further refine our ideas. We are excited about the concept and eager to use our skills to bring it to life. During the proposal presentation, Dr. Yiin provided additional feedback that we look forward to implementing. We will continue conducting market research and polling potential customers to gather their thoughts on the product. Finally, we will begin evaluating feasibility, exploring potential hardware options, and considering ways to expand the product with new features.

Overall, we enjoyed the opportunity to create a product that could have a real impact on the people it is intended to help and value the hands-on experience this course provides.